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Notifications 

5. Leaky tank

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Leaky tank (External resource) (1.0 points possible)

Leaky tank

We now consider a slightly different system. Suppose that tank 1 contains some volume of one fluid and tank 2 contains the same volume of another fluid. These two tanks are connected to the mixing tank (tank 3) of equal width. Tank 2 is initially empty. We set $a_{12} = a_{23} = 1$ and $a_{13} = 0$. As we showed in the previous problem, let the system run, then after a while we will end up with a mixture of the two fluids in tank 2, and the amount of each mixture will be one third of the total of the initial volumes of tank 1 and 3.

Unfortunately, tank 1 has a leak in it! This means we must adjust the equation for h_1 , to include a leakage term $Q_{leak} = -\mu h_1$ with $\mu > 0$, so that the system becomes:

$$\begin{aligned} \frac{dh_1}{dt} &= (h_2 - h_1) - \mu h_1, \\ \frac{dh_2}{dt} &= (h_3 - h_2) + (h_1 - h_2), \\ \frac{dh_3}{dt} &= (h_2 - h_3). \end{aligned}$$

We can then recast this in matrix form as

$$\dot{\mathbf{x}} = \begin{bmatrix} -1 - \mu & 1 & 0 \\ 1 & -2 & 1 \\ 0 & 1 & -1 \end{bmatrix} \mathbf{x}$$

with $\mathbf{x}$ defined as before. Let the initial conditions be

$$\mathbf{x}(0) = \begin{bmatrix} 1/2 \\ 0 \\ 1/2 \end{bmatrix}.$$

We know that the solution can be written in the form

$$\mathbf{x}(t) = c_1 e^{\lambda_1 t} \mathbf{v}_1 + c_2 e^{\lambda_2 t} \mathbf{v}_2 + c_3 e^{\lambda_3 t} \mathbf{v}_3.$$

Calculate the $\lambda_i, c_i, \mathbf{v}_i$ using MATLAB, and plot the solutions. Take $\mu = 1$.

Script

Save Reset

MATLAB Documentation (<https://www.mathworks.com/help/>)

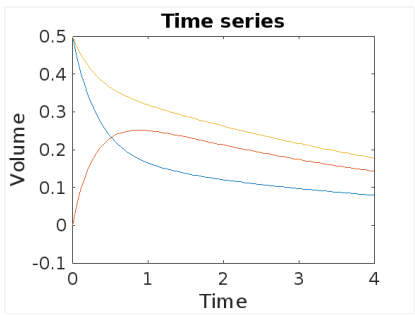
```
1 % Input the 3x3 matrix describing your linear system as a variable A
2 A = [-2 1 0; 1 -2 1; 0 1 -1];
3 % Input your initial conditions as a column vector x0
4 x0 = [1/2;0;1/2];
5 % Use eig(A) to find the eigenvalues and eigenvectors of A
6 % Define the eigenvectors as column vectors v1, v2 and v3,
7 % and the eigenvalues as lambda1, lambda2, lambda3
8 % so that A*v1 = lambda1*v1, etc.
9 [V,D] = eig(A);
10 v1 = V(:,1);
11 v2 = V(:,2);
12 v3 = V(:,3);
13 lambda1 = D(1,1);
14 lambda2 = D(2,2);
15 lambda3 = D(3,3);
16 % Calculate the column vector c = [c1;c2;c3] from the initial conditions using
17 c = inv(V)*x0;
18 c1 = c(1,1); c2 = c(2,1); c3 = c(3,1);
19 %% Define a row vector t with 100 equally spaced entries,
20 %% beginning with 0 and ending at 4.
21 t = linspace(0,4,100);
22 %% Define three row vectors h1, h2 and h3, with entries corresponding
23 %% to h1(t), h2(t) and h3(t) evaluated at each time in t.
24 h1 = c1*exp(lambda1*t)*v1(1) + c2*exp(lambda2*t)*v2(1) + c3*exp(lambda3*t)*v3(1);
25 h2 = c1*exp(lambda1*t)*v1(2) + c2*exp(lambda2*t)*v2(2) + c3*exp(lambda3*t)*v3(2);
26 h3 = c1*exp(lambda1*t)*v1(3) + c2*exp(lambda2*t)*v2(3) + c3*exp(lambda3*t)*v3(3);
27 %% Now use plot to plot the three vectors against time on the same figure
28 plot(t, h1);
29 hold on;
```

```
29 hold on
30 plot(t, h2);
31 plot(t, h3);
32 set(gca,'fontsize',18)
33 xlabel('Time')
34 ylabel('Volume')
35 title('Time series')
```

▶ Run Script



Output



5. Leaky tank

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